

BRIDGING THE GAP: ADDITIVE STRUCTURE IN THE TWIN PRIME COFACTORS

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ABSTRACT. We introduce the Bridging Conjecture: for every K , there exist $a, b, k \in A002822$ with $a < K$, $b < K$, and $k = a + b > K$, where $A002822$ denotes the set of positive integers m such that $6m - 1$ and $6m + 1$ are both prime (twin prime cofactors). We show that the Bridging Conjecture is equivalent to the Twin Prime Conjecture assuming Dubner's Conjecture 1, and that it follows unconditionally from Dubner's Conjecture 3. As a corollary, the Bridging Conjecture implies $\pi_2(x) \gg \log x$, a conditional lower bound on the twin prime counting function derived purely from the additive structure of $A002822$. We also re-express Dubner's exception set A243956 in set-theoretic terms as $D = \mathbb{N}_{\geq 1} \setminus \{i + j : i, j \in A002822\}$, which explains the additive origin of its elements rather than merely listing them, and reformulates Dubner's Conjecture 3 as the assertion that D is finite.

1. INTRODUCTION

A *twin prime pair* is a pair of primes $(p, p + 2)$. The integer $p + 1$ between them is divisible by 6; write $p + 1 = 6m$, so m is a positive integer with $6m - 1$ and $6m + 1$ both prime. The set of such cofactors,

$$A002822 = \{m \in \mathbb{N} : 6m - 1 \text{ and } 6m + 1 \text{ are both prime}\},$$

is OEIS sequence A002822 [8]. The Twin Prime Conjecture asserts that $A002822$ is infinite. A prime belonging to a twin prime pair is called a *t-prime*.

In 1996, Stephen Wagler communicated a question to Paulo Ribenboim: what is the smallest middle number that is not the sum of two t-primes? Ribenboim asked Harvey Dubner to investigate [2, pp. 291–299]. Dubner's paper [1] is the primary published source; he formulated four conjectures in response, which we recall in Section 2. Wagler's question is closest in spirit to Conjecture 2, which asks whether every middle number greater than 6 is the sum of two t-primes. In Section 3 we re-express Dubner's exception set A243956 in set-theoretic terms as the complement of the sumset of $A002822$ in $\mathbb{N}_{\geq 1}$, which explains the

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additive origin of the exceptions rather than merely listing them, and reformulates Dubner's Conjecture 3 as the assertion that this complement is finite. In Section 4 we introduce the *Bridging Conjecture* and establish its relationships to the Twin Prime Conjecture and Dubner's conjectures.

2. DUBNER'S CONJECTURES

Dubner [1] stated four conjectures, which we number as he does.

Conjecture 2.1 (Dubner, Conjecture 1). Every middle number greater than 6 is the sum of two middle numbers.

In terms of $A002822$: a middle number $6m$ is the sum of two middle numbers $6a$ and $6b$ if and only if $m = a + b$ with $a, b \in A002822$. So Conjecture 2.1 is equivalent to: every $m \in A002822$ with $m > 1$ is the sum of two smaller elements of $A002822$. Dubner verified this up to 2×10^{10} .

Conjecture 2.2 (Dubner, Conjecture 2). Every middle number greater than 6 is the sum of two t-primes.

Conjecture 2.3 (Dubner, Conjecture 3). Every multiple of 6 greater than 4206 is the sum of two middle numbers.

Conjecture 2.4 (Dubner, Conjecture 4a). Every even number greater than 4208 is the sum of two t-primes.

Conjecture 2.3 is strictly stronger than Conjecture 2.1: it requires all sufficiently large multiples of 6 to decompose, not only those that are middle numbers. The sumset reformulation of Conjecture 2.3 is developed in Section 3.

3. THE SUMSET OF $A002822$ AND DUBNER'S EXCEPTION SET

The natural object here is the sumset of $A002822$ with itself. For any $i, j \in A002822$ the sum $i + j$ is a positive integer, and this sum need not itself belong to $A002822$: it may lie in $A002822$ (a middle number decomposition of $i + j$) or in $A067611$ (the complement). In either case $i + j$ is an element of $\mathbb{N}_{\geq 1}$ produced by the pair (i, j) . Iterating over all pairs in $A002822$ and collecting their sums therefore generates a subset of $\mathbb{N}_{\geq 1}$ with no restriction on whether the sums are twin-prime cofactors. Let D denote the complement of this sumset in $\mathbb{N}_{\geq 1}$, i.e. the set of positive integers admitting no such representation; this is OEIS A243956 [9]. Dubner identified eleven multiples of 6 that are not the sum of two middle numbers,

$$\{96, 402, 516, 786, 906, 1116, 1146, 1266, 1356, 3246, 4206\},$$

and conjectured these are the only exceptions. A243956 is derived from Dubner's set by dividing each element by 6 to recover the cofactor and adjoining 1 (which lies in A002822 but cannot be expressed as a sum of two elements of A002822), giving

$$D = \{1, 16, 67, 86, 131, 151, 186, 191, 211, 226, 541, 701\}$$

up to 2×10^{10} , and Conjecture 2.3 states that D is exactly this finite set. Conjecture 2.3 is phrased in terms of multiples of 6 and the threshold 4206 because Dubner identified the exceptions by exhaustive search and could only describe them by listing them. The sumset of A002822 provides an intrinsic characterisation: an integer belongs to D if and only if it is not reachable from A002822 by addition, with no prior knowledge of D required.

Definition 3.1. $D = \mathbb{N}_{\geq 1} \setminus \{i + j : i, j \in A002822\}$, the complement of the sumset of A002822 in $\mathbb{N}_{\geq 1}$.

By definition, D and $\{i + j : i, j \in A002822\}$ partition $\mathbb{N}_{\geq 1}$. The known value of D up to 2×10^{10} is A243956 [9].

Conjecture 3.2 (Conjecture 2.3, sumset form). The complement of $\{i + j : i, j \in A002822\}$ in $\mathbb{N}_{\geq 1}$ is finite; equivalently, D is finite.

Let T denote the set of t-primes (OEIS A007534 [10]), i.e. primes belonging to a twin prime pair. Every sum of two t-primes is even. Analogously, Conjecture 2.4 has the following sumset reformulation.

Conjecture 3.3 (Conjecture 2.4, sumset form). The complement of $\{p + q : p, q \in T\}$ in the positive even integers is finite.

Remark 3.4. Dubner's computation confirms that every element of D greater than 1 lies in A067611: none of the eleven exceptions beyond 1 is a twin-prime cofactor. The element 1 lies in A002822 (since 5 and 7 are prime) but belongs to D because no two elements of A002822 sum to 1. Once A002822 is given, D is determined by its additive structure alone, without further reference to primality. Note also that Definition 3.1 is robust to the discovery of new exceptions: a new element of D beyond 701 would falsify Conjecture 2.3 (which fixes D as the known finite set) but would merely enlarge D in Definition 3.1, leaving the partition intact.

4. THE BRIDGING CONJECTURE

Conjecture 4.1 (Bridging Conjecture). For every $K \in \mathbb{N}$, there exist $a, b, k \in A002822$ with $a < K$, $b < K$, and $k = a + b > K$.

That is, one can always find two elements of $A002822$ below K whose sum is also in $A002822$ and exceeds K .

Theorem 4.2. *Conjecture 2.3 implies Conjecture 4.1, assuming the Twin Prime Conjecture.*

Proof. Given K , let k be the smallest element of $A002822$ greater than K (which exists by the Twin Prime Conjecture). For large enough K , $6k > 4206$, so Conjecture 2.3 gives $k = a + b$ with $a, b \in A002822$. By minimality of k , both a and b are at most K ; since $a + b = k > K$ they cannot both equal K , so $a, b < K$. $\square$

Theorem 4.3. *Conjecture 2.1 and the Twin Prime Conjecture together imply Conjecture 4.1.*

Proof. Given K , let k be the smallest element of $A002822$ greater than K . By Conjecture 2.1, $k = a + b$ for some $a, b \in A002822$ with $a, b < k$. By minimality of k , we have $a, b \leq K$, and since $a + b > K$ they cannot both equal K , so $a, b < K$. $\square$

Theorem 4.4. *Conjecture 4.1 implies the Twin Prime Conjecture.*

Proof. For any K , the Bridging Conjecture gives $k \in A002822$ with $k > K$. Since K is arbitrary, $A002822$ is infinite. $\square$

Corollary 4.5. *Assuming Conjecture 2.1,*

$$\text{Conjecture 4.1} \iff \text{Twin Prime Conjecture.}$$

Remark 4.6. The role of Conjecture 2.1 in Corollary 4.5 is asymmetric. The forward direction (Bridging Conjecture $\Rightarrow$ Twin Prime Conjecture) holds unconditionally: for any K , the Bridging Conjecture produces $k \in A002822$ with $k > K$, so $A002822$ is unbounded (Theorem 4.4). The reverse direction requires Conjecture 2.1: given that $A002822$ is infinite, let k be the smallest element above K ; Conjecture 2.1 decomposes $k = a + b$ with $a, b \in A002822$, and minimality of k forces $a, b < K$. Without Conjecture 2.1, the Twin Prime Conjecture alone does not guarantee that the next element above K is reachable as a sum of two smaller elements of $A002822$, and the reverse direction fails.

Corollary 4.7. *Conjecture 4.1 implies that for every $K \in \mathbb{N}$ there exists $k \in A002822$ with $K < k < 2K$. Equivalently, consecutive elements $m_n < m_{n+1}$ of $A002822$ satisfy $m_{n+1} < 2m_n$. Consequently, the counting function $\pi_2(x) = |\{m \in A002822 : m \leq x\}|$ satisfies $\pi_2(x) \gg \log x$.*

Proof. The Bridging Conjecture gives $a, b \in A002822$ with $a, b < K$ and $k = a + b \in A002822$ with $k > K$. Since $a, b < K$, we have $k = a + b < 2K$. The gap bound $m_{n+1} < 2m_n$ follows immediately. Since every interval $(m, 2m)$ contains an element of $A002822$, applying this inductively from $m_1 = 1$ gives at least $\lfloor \log_2 x \rfloor$ elements up to x , so $\pi_2(x) \gg \log x$. $\square$

Remark 4.8. Conjecture 2.1 is not required for Corollary 4.7: the bound $k < 2K$ is an immediate consequence of the Bridging Conjecture’s statement alone. What Conjecture 2.1 adds is qualitative: it asserts that *every* element of $A002822$ beyond 1 is reachable as a sum of two smaller elements of $A002822$, not only the specific k produced by the Bridging Conjecture. Note that sums of pairs from $A002822$ need not land in $A002822$ — many land in $A067611$ instead — so this is a non-trivial demand on the self-sumset structure of $A002822$. The upshot is that no element of $A002822$ beyond 1 is an additive dead-end: each one is witnessed by a pair of smaller elements.

5. DISCUSSION

Reformulation of Dubner’s conjectures. Dubner’s original statements of Conjectures 2.3 and 2.4 are phrased in terms of explicit numerical thresholds (4206 and 4208) derived from exhaustive computation. We reformulate them in Conjectures 3.2 and 3.3 as assertions that the complements of the relevant sumsets are finite. This rephrasing has two advantages. First, it removes the dependence on specific threshold values, which are artefacts of how far the computation was carried rather than features of the underlying mathematics. Second, it places Conjectures 2.3 and 2.4 in the same conceptual frame as Definition 3.1: all three are statements about the complement of a sumset in $\mathbb{N}_{\geq 1}$, making the structural parallel between them explicit.

The Bridging Conjecture as a splitting strategy. The Bridging Conjecture offers a way to split the problem. Rather than proving Conjecture 2.3 directly — which requires all sufficiently large multiples of 6 to decompose as sums of two middle numbers — one may instead attempt to establish the two weaker statements independently: Conjecture 2.1 (middle numbers themselves decompose) and the Bridging Conjecture ($A002822$ is unbounded, witnessed by sums of pairs of its own elements). Each of these makes a more focused demand on the additive structure of $A002822$, and either might be more tractable individually than Conjecture 2.3 in full generality.

It is worth noting that Conjecture 2.1 alone does not imply the Twin Prime Conjecture: $A002822$ could be finite with every element (beyond the smallest) expressible as a sum of two smaller elements, and Conjecture 2.1 would still hold vacuously beyond that point. The Bridging Conjecture supplies exactly what is missing: it asserts that for every K , some pair of elements of $A002822$ below K sums to an element of $A002822$ above K , forcing $A002822$ to be unbounded. In this sense the Bridging Conjecture can be read as a bootstrap principle: if $A002822$ is known to be non-empty up to K , it certifies the existence of a further element beyond K .

Relationship to D and Conjecture 2.3. The Bridging Conjecture is also strictly weaker than Conjecture 2.3 in a second respect. Conjecture 2.3 makes a precise global claim: D is exactly the finite set A243956. The Bridging Conjecture makes no claim about the size or composition of D — it requires only that the sumset $\{i + j : i, j \in A002822\}$ is unbounded, which is independent of whether D is finite. It does, however, place one indirect constraint: since the Bridging Conjecture requires infinitely many elements of $A002822$ to appear as sums of pairs of smaller elements, D cannot eventually absorb all positive integers.

Conditional lower bound on twin prime density. Corollary 4.7 gives a conditional lower bound on the twin prime counting function. The gap bound $m_{n+1} < 2m_n$ implies $\pi_2(x) \gg \log x$ assuming the Bridging Conjecture. For comparison, the best unconditional upper bound is Brun’s theorem [4], which gives $\pi_2(x) \ll x(\log \log x)^2/(\log x)^2$ via sieve methods; this does not establish infinitude. The Hardy–Littlewood conjecture [5] predicts the asymptotic $\pi_2(x) \sim 2C_2 x/(\log x)^2$ where $C_2 \approx 0.6601$ is the twin prime constant, but this remains unproven. Recent unconditional work of Zhang [6] and Maynard [7] establishes infinitely many prime pairs with bounded gaps, but does not yield an explicit lower bound on $\pi_2(x)$. The bound $\pi_2(x) \gg \log x$ from Corollary 4.7 is very weak — far below the Hardy–Littlewood prediction — but it is a consequence of a purely additive conjecture about $A002822$, with no direct appeal to sieve methods or the distribution of primes in progressions.

APPENDIX A. HEURISTIC EVIDENCE

The results of this paper are conditional on unproven conjectures. We present here heuristic arguments and computational evidence suggesting that both Conjecture 2.1 and the Bridging Conjecture are true. These arguments are informal and do not constitute proofs.

Evidence for the Bridging Conjecture. The Bridging Conjecture requires that for every K , some pair of elements of $A002822$ below K sums to an element of $A002822$ above K . By Corollary 4.7, this is equivalent to the gap bound $m_{n+1} < 2m_n$, or equivalently $\text{gap}(n)/m_{n+1} < 1/2$, where $\text{gap}(n) = m_{n+1} - m_n$.

Figure 1 plots $\text{gap}(n)/m_{n+1}$ for elements of $A002822$ up to $n = 10,000$, on a logarithmic scale. The dashed line marks the bound $1/2$. All observed ratios lie well below this bound, and the rolling mean decays steadily toward zero as m grows.

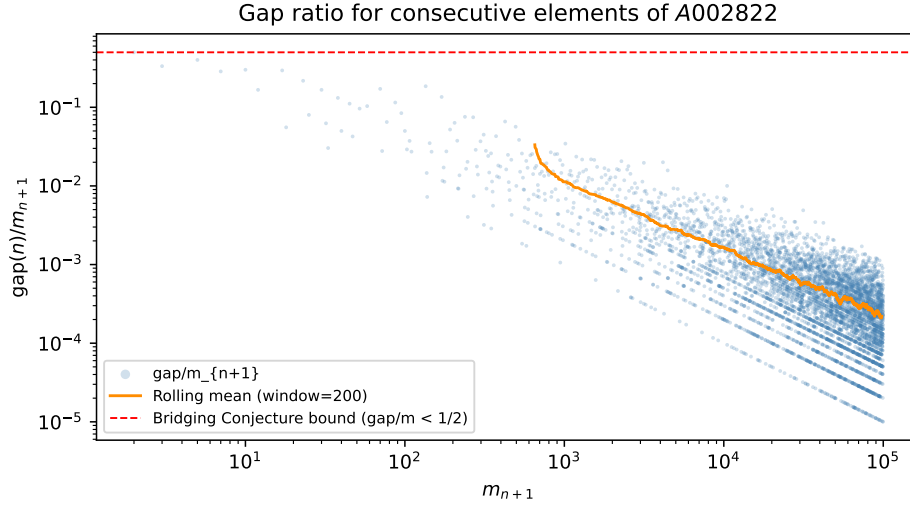


FIGURE 1. Gap ratio $\text{gap}(n)/m_{n+1}$ for consecutive elements of $A002822$ (log scale). The dashed line at $1/2$ marks the bound equivalent to the Bridging Conjecture. The rolling mean decays toward zero, consistent with the Hardy–Littlewood prediction.

The heuristic explanation is as follows. Under the Hardy–Littlewood conjecture [5], the density of twin prime cofactors near m is approximately $2C_2/(\log 6m)^2$, so the expected gap between consecutive elements of $A002822$ near m is $\sim (\log 6m)^2/2C_2$. Since $(\log 6m)^2 = o(m)$, the ratio $\text{gap}(n)/m_{n+1} \rightarrow 0$. This is far stronger than the bound of

$1/2$ required by the Bridging Conjecture, and explains the rapid decay seen in Figure 1. There is no heuristic mechanism by which this ratio would reverse its decay and approach $1/2$: that would require an anomalous and sustained absence of twin primes over intervals comparable in length to the primes themselves, which is inconsistent with the Hardy–Littlewood prediction.

Evidence for Conjecture 2.1. Conjecture 2.1 requires that every element of $A002822$ greater than 1 can be written as a sum of two smaller elements of $A002822$. Equivalently, the number of middle number decompositions of m is at least 1 for all $m \in A002822$ with $m > 1$.

Figure 2 plots this decomposition count for each $m \in A002822$ up to $n = 10,000$. The count is never zero for $m > 1$, and grows steadily with m . The dashed reference curve $c \cdot m/(\log m)^2$ tracks the rolling mean closely, consistent with the Hardy–Littlewood analogy.

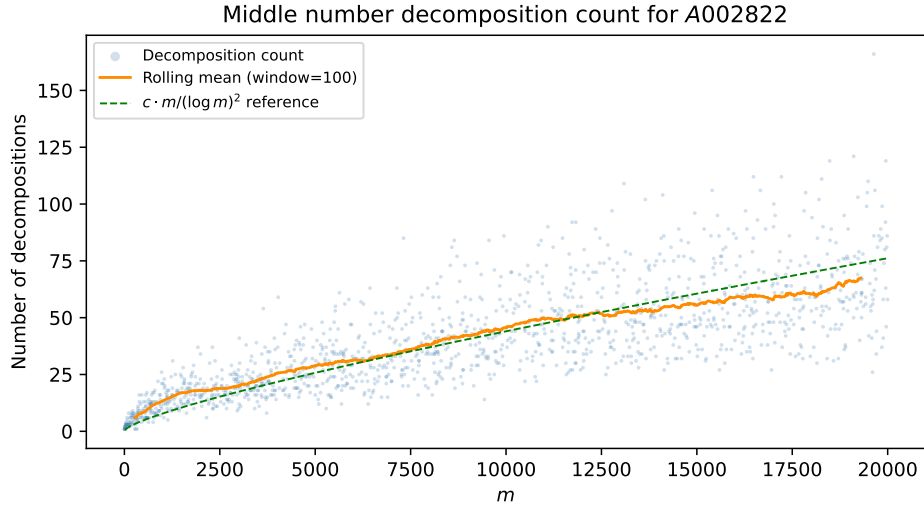


FIGURE 2. Number of ways to write $m \in A002822$ as a sum of two smaller elements of $A002822$, as a function of m . The count is never zero for $m > 1$. The dashed curve $c \cdot m/(\log m)^2$ is a Hardy–Littlewood analogy scaled to fit the data.

The heuristic argument is by analogy with the Goldbach problem. The number of ways to write an even integer n as a sum of two primes is expected to grow as $\sim C \cdot n/(\log n)^2$ by the Hardy–Littlewood circle method. The analogous count for $A002822$ replaces primes with twin prime cofactors: the number of pairs (a, b) with $a + b = m$ and $a, b \in$

A002822 is expected to grow at a similar rate, with a different constant reflecting the additional primality constraints on $6a \pm 1$ and $6b \pm 1$. Since this count grows without bound, it is in particular never permanently zero, supporting Conjecture 2.1.

Together, these two plots suggest that both conjectures hold comfortably in the computed range, and that the underlying heuristics give no reason to expect failure at larger values.

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